

WASP-12b

R-1916

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_151229 · created 2026-08-06T00:08:48+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457384.86844 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.022
Transit depth (Rp/Rs)^2	1.33 +/- 0.52 [%]
Orbital Inclination (inc)	76.98 +/- 2.92
Airmass coefficient 1 (a1)	1.066 +/- 0.016
Airmass coefficient 2 (a2)	-0.052 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.5 %
Best Comparison Star	#2 - [177, 218]
Optimal Aperture	3.78
Optimal Annulus	9.9
Transit Duration (day)	0.103 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.115 ± 0.022 vs 0.1178 (deviation 0.12σ)
Duration fit vs published	0.103 d vs 0.125 d (deviation 1.36σ)
Mid-time O–C	-1571.8 min
Depth SNR	4.9
Residual scatter	1.5 %

Light curve

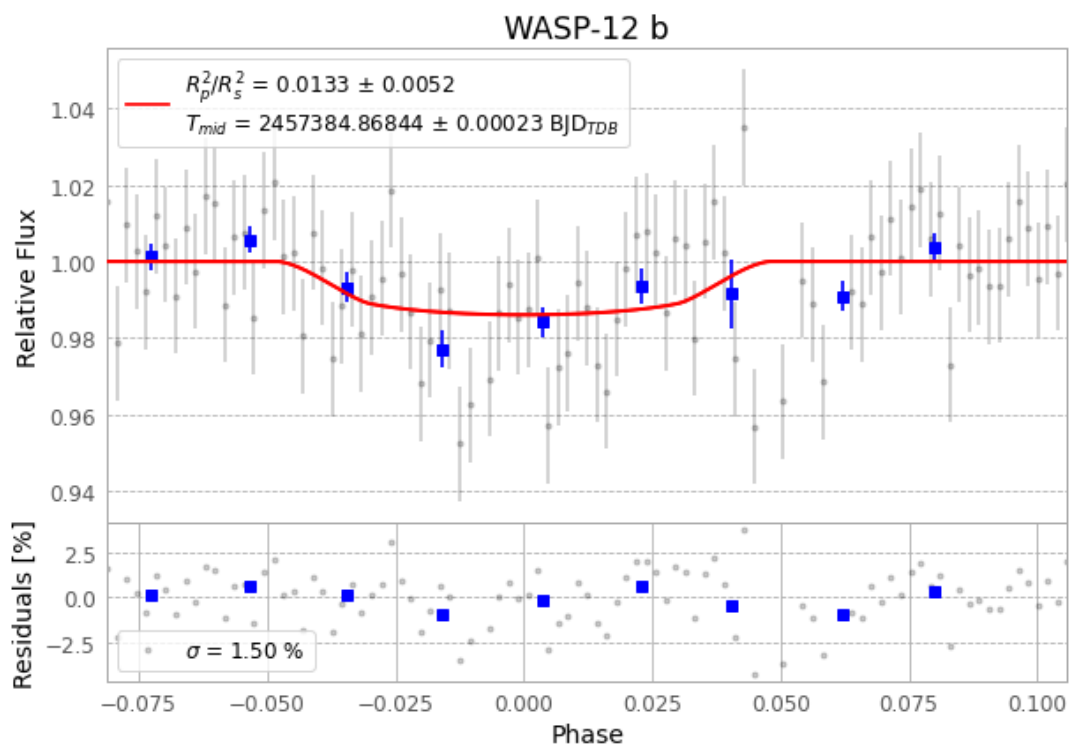


Figure 1 — FinalLightCurve_WASP-12 b_2015-12-27.png (SHA-256 8473757506b4bbee...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [288, 234], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 fb97b80ad70b4cc9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

99 FITS frames (65 MB), WASP-12151228063312.FITS → WASP-12151228112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-151229.log (7de151f89de0...), FinalLightCurve_WASP-12 b_2015-12-27.png (8473757506b4...), FinalLightCurve_WASP-12 b_2015-12-27.csv (925f017c5d64...)

Compute resources

Wall time 178.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 00:08Z.